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APPLIED PROBABILITY LABORATORY (22MDC25)

PROJECT 7

A MINI PROJECT ON APPLICATION OF JOINT PROBABILITY IN ANALYSING WEATHER DATA

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Task 1

1.Joint Probability Mass Function (Joint PMF)

The joint PMF describes the probability that two discrete random variables X and Y take on specific values at the same time. It gives a complete description of the relationship between the two variables.

$p(x, y) = P(X = x \text{ and } Y = y)$

$p(x, y) \geq 0$ for all values of x and y

$\Sigma x\ \Sigma y\ p(x, y) = 1 \rightarrow$ (Total probability is 1)

2.Marginal Probability Mass Function (Marginal PMF)

The marginal PMF gives the probability of one variable regardless of the other. It is obtained by summing the joint PMF over the other variable. This helps in understanding the behavior of a single variable when you have a joint distribution.

For variable X:

$p\_X(x) = \Sigma y\ p(x, y)$

For variable Y:

$p\_Y(y) = \Sigma x\ p(x, y)$

3.Conditional Expectation

The conditional expectation of X given Y=y is the average value of X, when we know that Y has a specific value y. It is a useful tool in probability and statistics to study how the expected value of one variable depends on another.

$E[X \mid Y = y] = \Sigma x\ x * P(X = x \mid Y = y)$

where $P(X = x \mid Y = y) = P(X = x \text{ and } Y = y) / P(Y = y)$

This is the conditional probability of X given Y=y , assuming P(Y=y)>0

4.Conditional Variance

The conditional variance of X given Y=y measures the spread or variability of XXX around its
conditional mean, E[X ∣ Y=y]E[X | Y = y]E[X ∣ Y=y]. It tells us how much X fluctuates when Y is fixed at a certain value.

$$Var(X \mid Y = y) = E[(X - E[X \mid Y = y])^2 \mid Y = y]$$
$$= \Sigma x\ (x - E[X \mid Y = y])^2 * P(X = x \mid Y = y)$$

Task 2

Go to Web Site http://www.indiawaterportal.org/met_data/.
Download following data

1. **Monthly rainfall data of Thanjavur District from 1901 to 2002**
2. **Monthly average maximum temperature at Thanjavur from 1901 to 2002**
3. **Monthly average minimum temperature at Thanjavur from 1901 to 2002**

Write a Script file in R to identify the month at which

1. **rainfall and maximum temperature are correlated to the maximum, that is correlation coefficient is maximum**
2. **rainfall and minimum temperature are correlated to the maximum, that is correlation coefficient is maximum**
3. **maximum temperature and minimum temperature are correlated to the maximum, that is correlation coefficient is maximum**

Source Code

```
# Load the datasets using base R
rainfall <- read.csv("C:\\R_Project\\rainfall.csv", header = TRUE)
max_temp <- read.csv("C:\\R_Project\\max_temp.csv", header = TRUE)
min_temp <- read.csv("C:\\R_Project\\min_temp.csv", header = TRUE)

# Remove the first column if it's 'Year' to keep only monthly data
rainfall_data <- rainfall[, -1]
max_temp_data <- max_temp[, -1]
min_temp_data <- min_temp[, -1]

# Get month names (column names)
months <- colnames(rainfall_data)

# Initialize vectors for storing correlation values
cor_rain_max <- numeric(length(months))
cor_rain_min <- numeric(length(months))
cor_max_min  <- numeric(length(months))
```

```
# Compute correlations month by month
for (i in seq_along(months)) {
 cor_rain_max[i] <- cor(rainfall_data[[i]], max_temp_data[[i]], use =
"complete.obs")
 cor_rain_min[i] <- cor(rainfall_data[[i]], min_temp_data[[i]], use =
"complete.obs")
 cor_max_min[i] <- cor(max_temp_data[[i]], min_temp_data[[i]], use =
"complete.obs")
}

# Find the month with maximum correlation
max_cor_rain_max <- months[which.max(cor_rain_max)]
max_cor_rain_min <- months[which.max(cor_rain_min)]
max_cor_max_min <- months[which.max(cor_max_min)]

# Print results
cat("Month with highest correlation between Rainfall and Max Temp:",
max_cor_rain_max, "\n")
cat("Month with highest correlation between Rainfall and Min Temp:",
max_cor_rain_min, "\n")
cat("Month with highest correlation between Max Temp and Min Temp:",
max_cor_max_min, "\n")
```

Output

```
> # Load the datasets using base R
> rainfall <- read.csv("C:\\R_Project\\rainfall.csv", header = TRUE)
> max_temp <- read.csv("C:\\R_Project\\max_temp.csv", header = TRUE)
> min_temp <- read.csv("C:\\R_Project\\min_temp.csv", header = TRUE)
>
> # Remove the first column if it's 'Year' to keep only monthly data
> rainfall_data <- rainfall[, -1]
> max_temp_data <- max_temp[, -1]
> min_temp_data <- min_temp[, -1]
>
> # Get month names (column names)
> months <- colnames(rainfall_data)
>
> # Initialize vectors for storing correlation values
> cor_rain_max <- numeric(length(months))
> cor_rain_min <- numeric(length(months))
> cor_max_min  <- numeric(length(months))
>
> # Compute correlations month by month
> for (i in seq_along(months)) {
+   cor_rain_max[i] <- cor(rainfall_data[[i]], max_temp_data[[i]], use = "complete.obs")
+   cor_rain_min[i] <- cor(rainfall_data[[i]], min_temp_data[[i]], use = "complete.obs")
+   cor_max_min[i]  <- cor(max_temp_data[[i]], min_temp_data[[i]], use = "complete.obs")
+ }
>
> # Find the month with maximum correlation
> max_cor_rain_max <- months[which.max(cor_rain_max)]
> max_cor_rain_min <- months[which.max(cor_rain_min)]
> max_cor_max_min  <- months[which.max(cor_max_min)]
>
> # Print results
> cat("Month with highest correlation between Rainfall and Max Temp:", max_cor_rain_max, "\n")
Month with highest correlation between Rainfall and Max Temp: Apr
> cat("Month with highest correlation between Rainfall and Min Temp:", max_cor_rain_min, "\n")
Month with highest correlation between Rainfall and Min Temp: Nov
> cat("Month with highest correlation between Max Temp and Min Temp:", max_cor_max_min, "\n")
Month with highest correlation between Max Temp and Min Temp: Nov
>
```

Task 3

Write a script file in R to form a discrete probability distribution between :

1. **rainfall and max temperature at the month where correlation coefficient is maximum**
2. **rainfall and min temperature at the month where correlation coefficient is maximum**
3. **max temperature and min temperature at the month where correlation coefficient is maximum.**

   [The joint probabilities can be worked out by dividing the data on X and Y into different bins (about 5 to 7) and then finding the frequencies in each cell]
4. **Compute the marginal p.m.fs, conditional expectations and conditional variances.**

Source Code

```
# Load datasets
rainfall <- read.csv("rainfall_thanjavur_1901_2002.csv")
max_temp <- read.csv("max_temperature_thanjavur_1901_2002.csv")
min_temp <- read.csv("min_temperature_thanjavur_1901_2002.csv")

# Remove Year column
rainfall_data <- rainfall[ , -1]
max_temp_data <- max_temp[ , -1]
min_temp_data <- min_temp[ , -1]

# Function to find month with highest correlation
get_max_corr_month <- function(df1, df2) {
  cors <- sapply(1:12, function(i) cor(df1[[i]], df2[[i]], use =
"complete.obs"))
  return(which.max(cors))  # returns month index (1-12)
}

# Find months with highest correlation
month_rm <- get_max_corr_month(rainfall_data, max_temp_data)
month_rn <- get_max_corr_month(rainfall_data, min_temp_data)
month_mn <- get_max_corr_month(max_temp_data, min_temp_data)
```

```r
# Discretize into bins
get_joint_distribution <- function(x, y, bins = 5) {
 x_cut <- cut(x, breaks = bins)
 y_cut <- cut(y, breaks = bins)
 joint_table <- table(x_cut, y_cut)
 joint_prob <- prop.table(joint_table) # joint probability matrix
 return(joint_prob)
}

# Compute marginal pmfs
get_marginals <- function(joint_prob) {
 x_marginal <- rowSums(joint_prob)
 y_marginal <- colSums(joint_prob)
 return(list(x = x_marginal, y = y_marginal))
}

# Compute conditional expectation and variance of Y given X
get_conditional_stats <- function(joint_prob) {
 x_levels <- rownames(joint_prob)
 y_levels <- colnames(joint_prob)

 # Midpoints of bins (approximated)
 x_vals <- sapply(strsplit(x_levels, ","), function(s) mean(as.numeric(gsub("
[^0-9.]", "", s))))
 y_vals <- sapply(strsplit(y_levels, ","), function(s) mean(as.numeric(gsub("
[^0-9.]", "", s))))

 cond_exp <- numeric(length(x_vals))
 cond_var <- numeric(length(x_vals))

 for (i in 1:length(x_vals)) {
 row_probs <- joint_prob[i, ]
 total_p <- sum(row_probs)
 if (total_p == 0) {
 cond_exp[i] <- NA
 cond_var[i] <- NA
 }
```

```
else {
 cond_y <- as.numeric(y_vals)
 mean_y <- sum(row_probs * cond_y) / total_p
 var_y <- sum(row_probs * (cond_y - mean_y)^2) / total_p
 cond_exp[i] <- mean_y
 cond_var[i] <- var_y
 }
 }
 return(data.frame(X_bin = x_levels, E_Y_given_X = cond_exp,
Var_Y_given_X = cond_var))
}

# Analysis for Rainfall vs Max Temp
cat("\n=== Rainfall vs Max Temp (Month:", month_rm, ")\n")
joint_rm <- get_joint_distribution(rainfall_data[[month_rm]],
max_temp_data[[month_rm]])
marginals_rm <- get_marginals(joint_rm)
stats_rm <- get_conditional_stats(joint_rm)
print(stats_rm)

# Analysis for Rainfall vs Min Temp
cat("\n=== Rainfall vs Min Temp (Month:", month_rn, ")\n")
joint_rn <- get_joint_distribution(rainfall_data[[month_rn]],
min_temp_data[[month_rn]])
marginals_rn <- get_marginals(joint_rn)
stats_rn <- get_conditional_stats(joint_rn)
print(stats_rn)

# Analysis for Max Temp vs Min Temp
cat("\n=== Max Temp vs Min Temp (Month:", month_mn, ")\n")
joint_mn <- get_joint_distribution(max_temp_data[[month_mn]],
min_temp_data[[month_mn]])
marginals_mn <- get_marginals(joint_mn)
stats_mn <- get_conditional_stats(joint_mn)
print(stats_mn)
```

Output

```
> # Load datasets
> rainfall <- read.csv("C:\\R_Project\\rainfall.csv")
> max_temp <- read.csv("C:\\R_Project\\max_temp.csv")
> min_temp <- read.csv("C:\\R_Project\\min_temp.csv")
>
> # Remove Year column
> rainfall_data <- rainfall[ , -1]
> max_temp_data <- max_temp[ , -1]
> min_temp_data <- min_temp[ , -1]
>
> # Function to find month with highest correlation
> get_max_corr_month <- function(df1, df2) {
+   cors <- sapply(1:12, function(i) cor(df1[[i]], df2[[i]], use = "complete.obs"))
+   return(which.max(cors))  # returns month index (1-12)
+ }
>
> # Find months with highest correlation
> month_rm <- get_max_corr_month(rainfall_data, max_temp_data)
> month_rn <- get_max_corr_month(rainfall_data, min_temp_data)
> month_mn <- get_max_corr_month(max_temp_data, min_temp_data)
>
> # Discretize into bins
> get_joint_distribution <- function(x, y, bins = 5) {
+   x_cut <- cut(x, breaks = bins)
+   y_cut <- cut(y, breaks = bins)
+   joint_table <- table(x_cut, y_cut)
+   joint_prob <- prop.table(joint_table)  # joint probability matrix
+   return(joint_prob)
+ }
>
> # Compute marginal pmfs
> get_marginals <- function(joint_prob) {
+   x_marginal <- rowSums(joint_prob)
+   y_marginal <- colSums(joint_prob)
+   return(list(x = x_marginal, y = y_marginal))
+ }
> # Compute conditional expectation and variance of Y given X
> get_conditional_stats <- function(joint_prob) {
+   x_levels <- rownames(joint_prob)
+   y_levels <- colnames(joint_prob)
+
+   # Midpoints of bins (approximated)
+   x_vals <- sapply(strsplit(x_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))
+   y_vals <- sapply(strsplit(y_levels, ","), function(s) mean(as.numeric(gsub("[^0-9.]", "", s))))
+
+   cond_exp <- numeric(length(x_vals))
+   cond_var <- numeric(length(x_vals))
+
+   for (i in 1:length(x_vals)) {
+     row_probs <- joint_prob[i, ]
+     total_p <- sum(row_probs)
+     if (total_p == 0) {
+       cond_exp[i] <- NA
+       cond_var[i] <- NA
+     } else {
+       cond_y <- as.numeric(y_vals)
+       mean_y <- sum(row_probs * cond_y) / total_p
+       var_y <- sum(row_probs * (cond_y - mean_y)^2) / total_p
+       cond_exp[i] <- mean_y
+       cond_var[i] <- var_y
+     }
+   }
+   return(data.frame(X_bin = x_levels, E_Y_given_X = cond_exp, Var_Y_given_X = cond_var))
+ }
>
> # Analysis for Rainfall vs Max Temp
> cat("\n=== Rainfall vs Max Temp (Month:", month_rm, ")\n")

=== Rainfall vs Max Temp (Month: 4 )
> joint_rm <- get_joint_distribution(rainfall_data[[month_rm]], max_temp_data[[month_rm]])
> marginals_rm <- get_marginals(joint_rm)
> stats_rm <- get_conditional_stats(joint_rm)
> print(stats_rm)
         X_bin E_Y_given_X Var_Y_given_X
1 (3.91,82.7]    34.79800     10.469496
2  (82.7,161]    37.33235      9.036453
3   (161,239]    36.23056     11.740594
4   (239,318]    35.65714      9.092925
5   (318,397]    35.76905      8.435113
>
```

```
> # Analysis for Rainfall vs Min Temp
> cat("\n=== Rainfall vs Min Temp (Month:", month_rn, ")\n")

=== Rainfall vs Min Temp (Month: 11 )
> joint_rn <- get_joint_distribution(rainfall_data[[month_rn]], min_temp_data[[month_rn]])
> marginals_rn <- get_marginals(joint_rn)
> stats_rn <- get_conditional_stats(joint_rn)
> print(stats_rn)
        X_bin E_Y_given_X Var_Y_given_X
1 (6.71,85.5]    22.45000      5.070000
2  (85.5,164]    21.88125      8.813086
3   (164,242]    22.96071     10.030064
4   (242,321]    23.51053      9.037521
5   (321,400]    23.60556      7.771914
>
> # Analysis for Max Temp vs Min Temp
> cat("\n=== Max Temp vs Min Temp (Month:", month_mn, ")\n")

=== Max Temp vs Min Temp (Month: 11 )
> joint_mn <- get_joint_distribution(max_temp_data[[month_mn]], min_temp_data[[month_mn]])
> marginals_mn <- get_marginals(joint_mn)
> stats_mn <- get_conditional_stats(joint_mn)
> print(stats_mn)
        X_bin E_Y_given_X Var_Y_given_X
1   (30,32.4]    22.07857      7.467041
2 (32.4,34.8]    22.91429     10.467653
3 (34.8,37.1]    22.65682      9.309840
4 (37.1,39.5]    23.56429      6.846224
5 (39.5,41.9]    23.32941      7.105017
>
```

Problem statement

1. Crop Failure and Farmer Distress

- Problem: Farmers rely heavily on predictable rainfall and temperature.
- When low rainfall and high temperature occur together (as your project identifies via joint probability), it leads to:
  - Drought conditions.
  - Withering crops.
  - Increased irrigation costs.
- Impact: Crop loss, farmer debts, and even farmer suicides in extreme cases.

2. Water Scarcity

- Problem: Reservoirs and water tanks are replenished by seasonal rain.
- If rainfall is low and evaporation (due to high temperature) is high during the same months:
  - Water levels drop quickly.
  - People face shortages in drinking water, irrigation, and household use.
- Example: Many towns in Tamil Nadu face water rationing during such months.

3. Heat-Related Health Risks

- Problem: High temperatures, especially when not offset by rainfall (which usually cools things down), cause:
  - Heat stroke.
  - Dehydration.
  - Increased hospital visits among elderly and children.
- Your project's conditional probability of temperature extremes can help predict such months.

Problem statement

4. Education Disruption

- Problem: During peak summer with little rain, schools (especially rural ones without fans) often:
    - Send children home early.
    - Reduce class hours.
- This affects learning outcomes, especially for economically disadvantaged students.

5. Power Outages and Grid Strain

- Problem: Hot, dry months increase power demand for fans and air conditioning.
- Simultaneously, reduced rainfall can decrease hydroelectric generation.
- Impact:
    - Blackouts in rural and semi-urban areas.
    - Interruptions to small businesses.

6. Urban Drainage and Flooding

- On the other end, if heavy rainfall and lower temperatures coincide:
    - Sudden downpours can flood streets and homes.
    - Poorly planned cities like parts of Thanjavur can experience drainage failure.
- Joint analysis helps city planners prepare for these specific patterns.

Proposed solution

1. Crop Failure and Farmer Distress

Proposed Solutions:

- Predictive Advisory System: Use joint probability models to forecast high-risk months (low rain + high temperature) and alert farmers early.
- Crop Diversification: Promote short-duration or drought-tolerant crops during high-risk months.
- Subsidized Micro-irrigation: Encourage drip irrigation to reduce water usage during dry-hot months.
- Weather Insurance Schemes: Expand government insurance linked to weather forecasts to compensate crop loss.

2. Water Scarcity

Proposed Solutions:

- Early Water Management Alerts: Use month-wise joint probability data to issue early warnings for low-rainfall + high-evaporation months.
- Rainwater Harvesting Campaigns: Mandate community rainwater harvesting units, especially in critical summer months.
- Reservoir Operation Planning: Time water release and storage in dams based on expected joint rainfall-temp patterns.
- Revival of Traditional Water Bodies: Restore ponds/tanks in villages which help retain water during bad rainfall years.

3. Heat-Related Health Risks

Proposed Solutions:

- Heat Action Plans: Government to pre-deploy water stations, mobile health clinics in months with high heat and low rain.
- Public Awareness Drives: Educate citizens to avoid peak sun exposure during predicted extreme months.
- School and Work Adjustments: Allow flexible school/work hours and siesta policies during forecasted “extreme heat + dry” months.

Proposed solution

4. Education Disruption
Proposed Solutions:

- **Infrastructure Improvements: Use forecast data to prioritize funding for fans, coolers, shaded areas in rural schools.**
- **Remote Learning Backup: Prepare recorded lessons or hybrid models for high-heat, low-rain months.**
- **Water Supply to Schools: Ensure school water tanks are filled in advance during expected dry spells.**

5. Power Outages and Grid Strain
Proposed Solutions:

- **Demand Forecasting: Use correlation data to anticipate summer peak loads and ramp up supply proactively.**
- **Distributed Solar Backup: Install solar panels in rural schools, homes, and government offices to reduce grid dependency.**
- **Load-Shedding Management: Design smarter, predictive load-shedding plans based on weather data.**

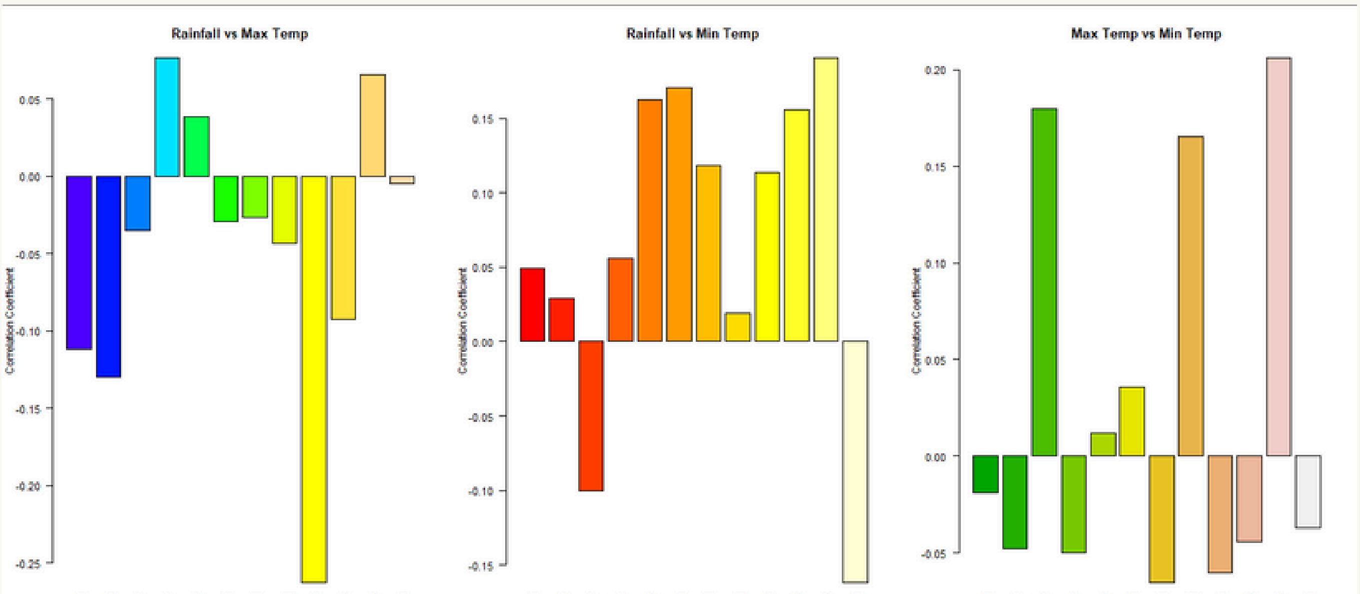
6. Urban Drainage and Flooding
Proposed Solutions:

- **Urban Rainfall Prediction: Use the project's joint analysis to predict sudden intense rainfall during cooler months.**
- **Drainage Mapping & Upgrades: Identify months/locations with recurring overflow risk and redesign drainage layouts accordingly.**
- **Community Preparedness: Train local bodies to use forecast tools and pre-emptively clean drains or relocate residents.**

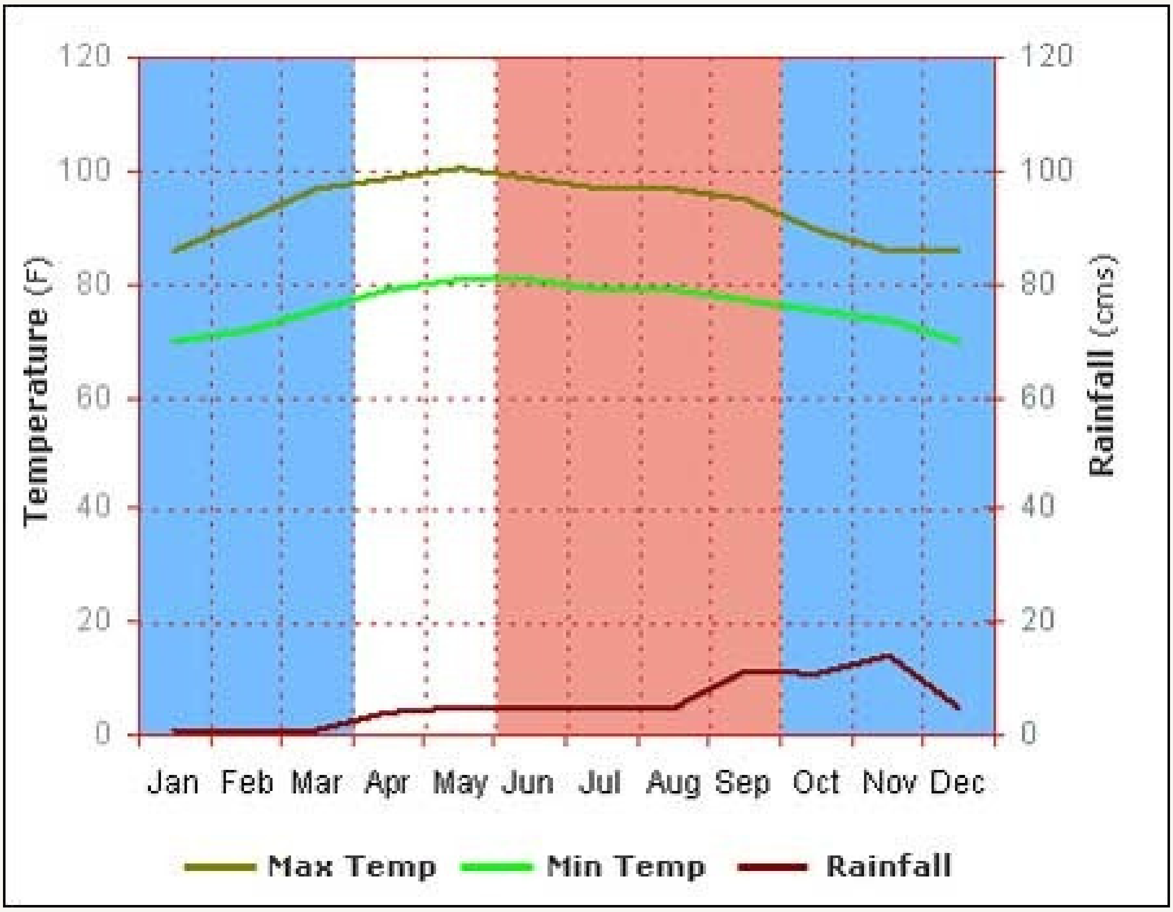
Visual graph plot

```
rainfall <- read.csv("D:/rainfall.csv", header = TRUE)
max_temp <- read.csv("D:/max_temp.csv", header = TRUE)
min_temp <- read.csv("D:/min_temp.csv", header = TRUE)
rainfall_data <- rainfall[, -1]
max_temp_data <- max_temp[, -1]
min_temp_data <- min_temp[, -1]
get_correlation_vector <- function(df1, df2) {
  sapply(1:12, function(i) cor(df1[[i]], df2[[i]], use =
"complete.obs"))
}
month_names <- colnames(rainfall_data)
cor_rm <- get_correlation_vector(rainfall_data, max_temp_data)
cor_rn <- get_correlation_vector(rainfall_data, min_temp_data)
cor_mn <- get_correlation_vector(max_temp_data,
min_temp_data)
par(mfrow = c(1, 3))
barplot(cor_rm,
        col = topo.colors(12),
        names.arg = month_names,
        las = 2,
        main = "Rainfall vs Max Temp",
        ylab = "Correlation Coefficient")
barplot(cor_rn,
        col = heat.colors(12),
        names.arg = month_names,
        las = 2,
        main = "Rainfall vs Min Temp",
        ylab = "Correlation Coefficient")
barplot(cor_mn,
        col = terrain.colors(12),
        names.arg = month_names,
        las = 2,
        main = "Max Temp vs Min Temp",
        ylab = "Correlation Coefficient")
```

Visual graph plot

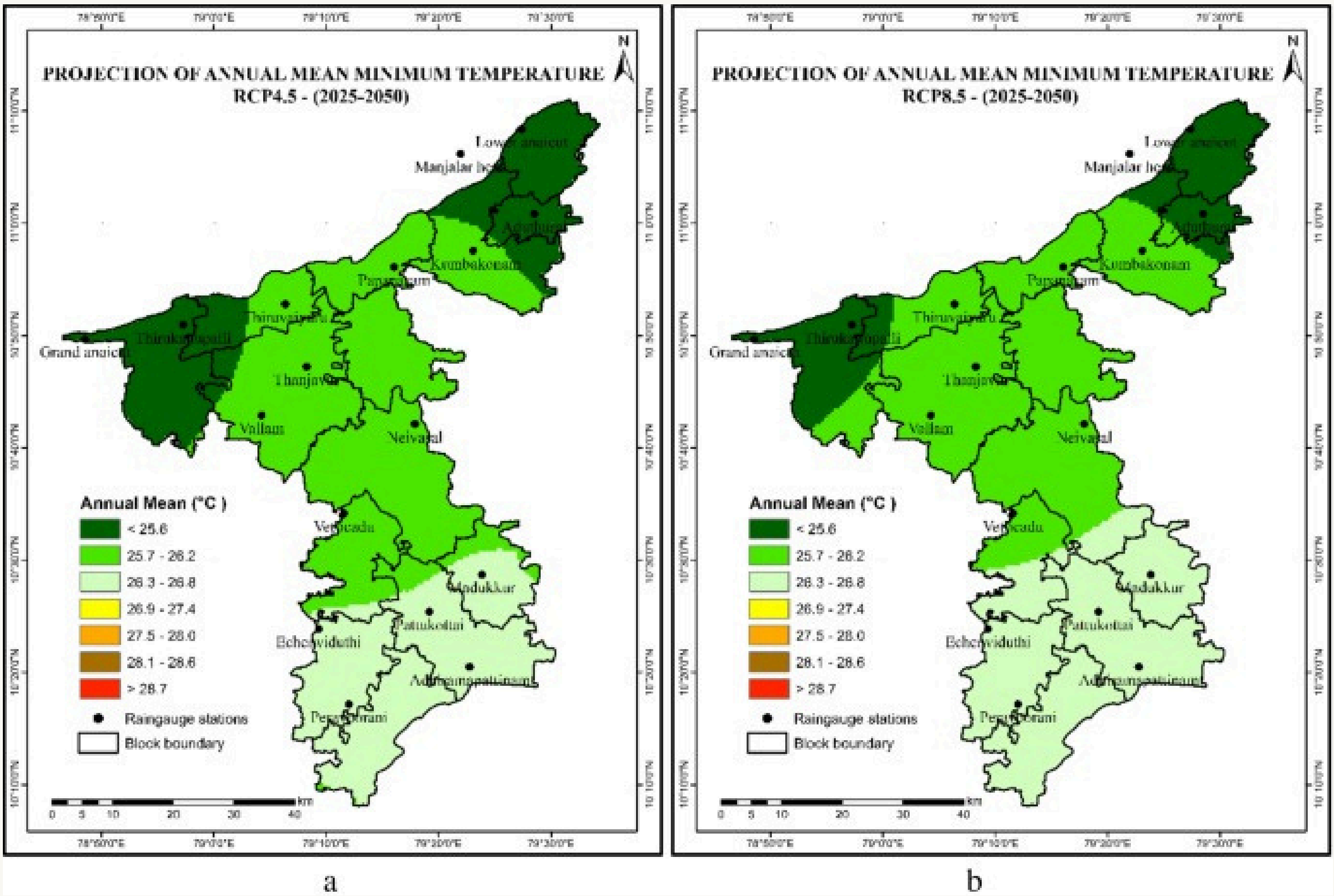


General graph



Future prediction

From 2025 to 2050.min and max temp in thanjavur :



Literature Review

The study of climate behavior through statistical methods has long been an area of interest, especially given the increasing unpredictability of weather patterns due to climate change. Joint probability analysis, a fundamental concept in probability theory, has proven to be a powerful tool in understanding the interdependence between multiple meteorological variables such as rainfall, maximum temperature, and minimum temperature.

1. Joint Probability and Weather Analytics

Several studies (e.g., Wilks, 1995; Katz & Brown, 1992) have explored the use of joint probability mass functions (PMFs) to model the relationship between weather parameters. These models help quantify the likelihood of simultaneous occurrences, such as low rainfall with high temperatures, which are critical conditions for identifying drought-prone periods. Joint PMFs, along with marginal and conditional distributions, allow for a multidimensional understanding of weather systems.

2. Correlation as a Predictive Tool

Pearson correlation has been widely employed to measure linear dependence between climate variables. Researchers like Hamed & Rao (1998) have used monthly and seasonal correlations to predict monsoon behavior in India. In this project, monthly correlation analyses have helped identify which months exhibit the strongest associations between rainfall and temperature parameters, a strategy also seen in regional studies by the Indian Meteorological Department (IMD).

3. Climate Studies in the Indian Context

India's diverse climatic zones and dependence on monsoons make it a unique case for probabilistic weather modeling. Specific to Tamil Nadu, studies by the India Water Portal and Tamil Nadu Agricultural University (TNAU) have documented long-term temperature and rainfall trends. The project's focus on Thanjavur aligns with these regional studies, providing localized insights from over a century of data.

4. Risk Modeling and Impact Assessment

Previous literature (e.g., IPCC Reports; World Bank Climate Resilience Reports) emphasizes the need for probabilistic modeling in managing agricultural risks and water resources. Conditional expectations and variances, as used in this study, are particularly valuable in estimating the impact of extreme conditions. This approach aligns with methodologies used in drought risk modeling and disaster management plans.

Conclusion

This project effectively demonstrates how joint probability and statistical analysis can be applied to real-world weather data to gain meaningful insights into climate behavior. By analyzing over a century's worth of rainfall and temperature data for Thanjavur, we identified the months with the strongest correlations between:

- Rainfall and maximum temperature
- Rainfall and minimum temperature
- Maximum and minimum temperature

Using joint probability mass functions (PMFs), we modeled the likelihood of specific rainfall-temperature combinations, and derived marginal PMFs, conditional expectations, and conditional variances. This enabled us to not only understand the relationship between these variables but also anticipate extreme weather patterns.

Such statistical models have significant practical implications:

- They help farmers prepare for crop risks due to drought or heat.
- Assist policymakers in water management and disaster preparedness.
- Provide early indicators for health and energy demand planning.

Ultimately, this project highlights the power of applied mathematics and data science in tackling climate-related challenges, proving that even historical datasets can inform modern solutions to pressing issues like agriculture sustainability, urban planning, and climate resilience.